Intro to ML and k nearest neighbours

Machine Learning, lecture 1

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YMLS Uzbekistan, fall 2026

YMLS



girafe ai

Today

1. What machine learning is
2. How an ML project works
3. Our first model: k nearest neighbours

Exit quiz in the last 10 minutes.

kNN moves from lecture 4 to lecture 1

Last year it came with SVM and PCA. Naive Bayes moves to lecture 2, and homework 1 is still kNN from scratch.

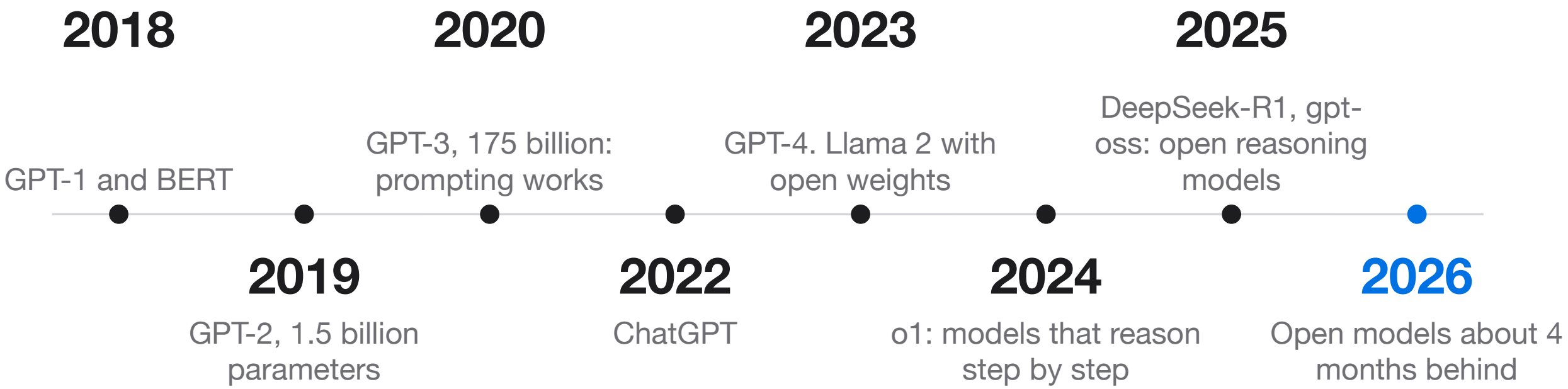
Machine learning today

A short history of AI



Not to scale.

Language models, 2018 to 2026



Open weights: anyone can download the model and run it.

Benchmarks fill up fast

Benchmark	What it tests	At launch	Nov 2026
MMLU, 2020	57 school and university subjects	44%	**90%+**
Humanity's Last Exam, 2025	2,500 expert questions	8%	**61%**
ARC-AGI-2, 2025	Puzzles that people find easy	under 10%	**95%**
ARC-AGI-3, 2026	Interactive puzzle games	under 1%	**63%**

Best AI score. Sources: Hendrycks et al. 2020, lastexam.ai, Artificial Analysis, arcprize.org.

Vision: detect, segment, generate



Games: AlphaGo and OpenAI Five



AlphaGo beat Lee Sedol 4 games to 1, 2016



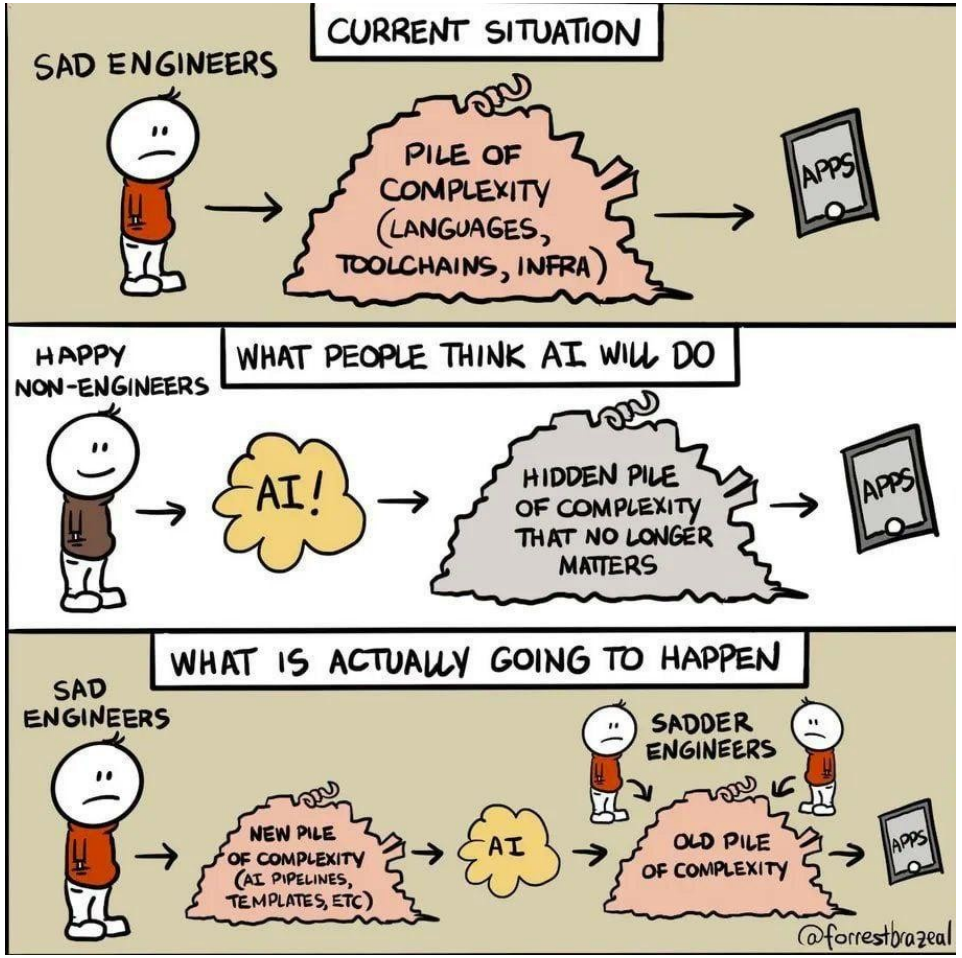
OpenAI Five beat the Dota 2 world champions, 2019

Two Nobel Prizes went to machine learning in 2024

Physics: Hopfield and Hinton, neural networks.

Chemistry: Baker, Hassabis and Jumper, protein design and AlphaFold.

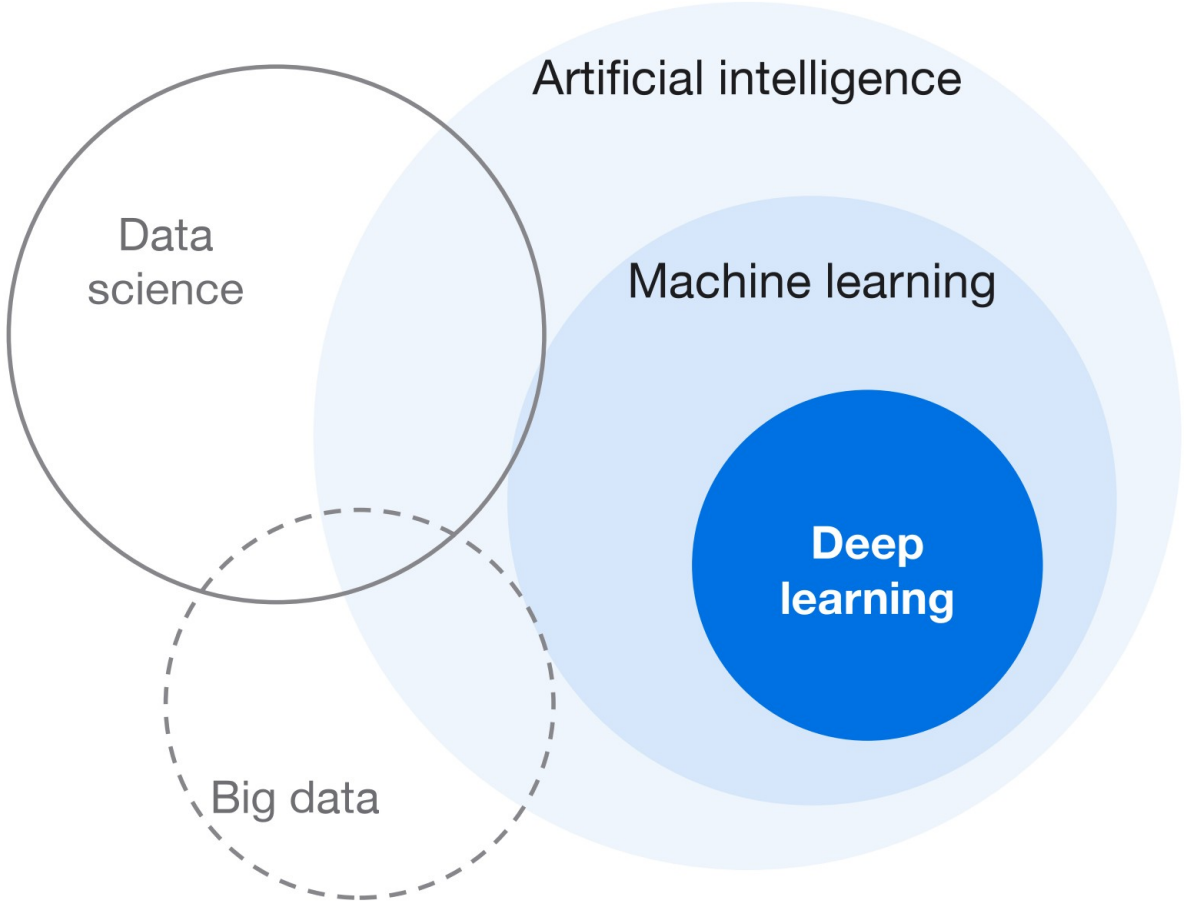
What are AI and ML for you?



Comic by Forrest Brazeal

What is machine learning?

Terms around ML



Artificial intelligence



Any system that perceives its environment and acts to reach its goals.

By this definition, your electric kettle is AI.

Software 1.0, 2.0, 3.0

1.0

code

A person writes the rules

2.0

weights

Rules are learned from data and answers

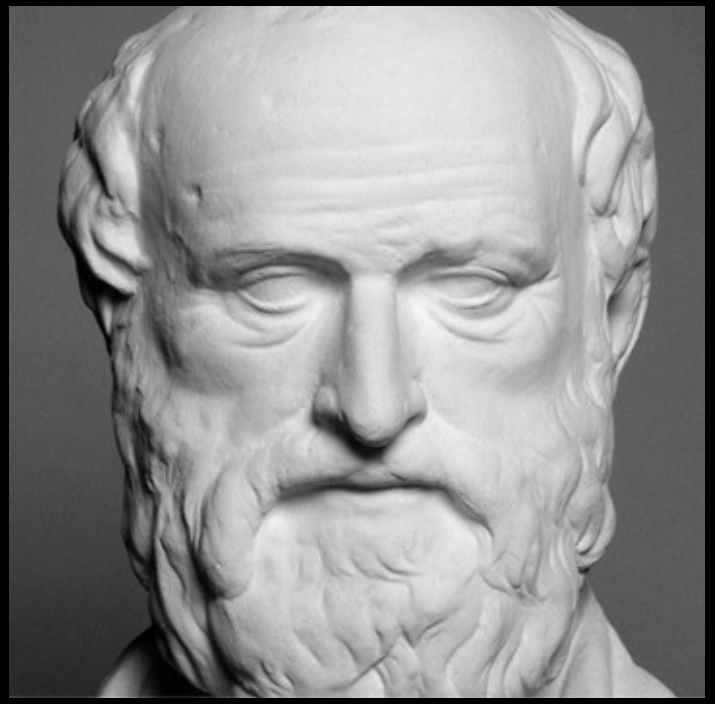
3.0

prompts

Plain-language instructions for an LLM

This course is about 2.0. Terms by Andrej Karpathy, 2017 and 2025.

Long before machine learning



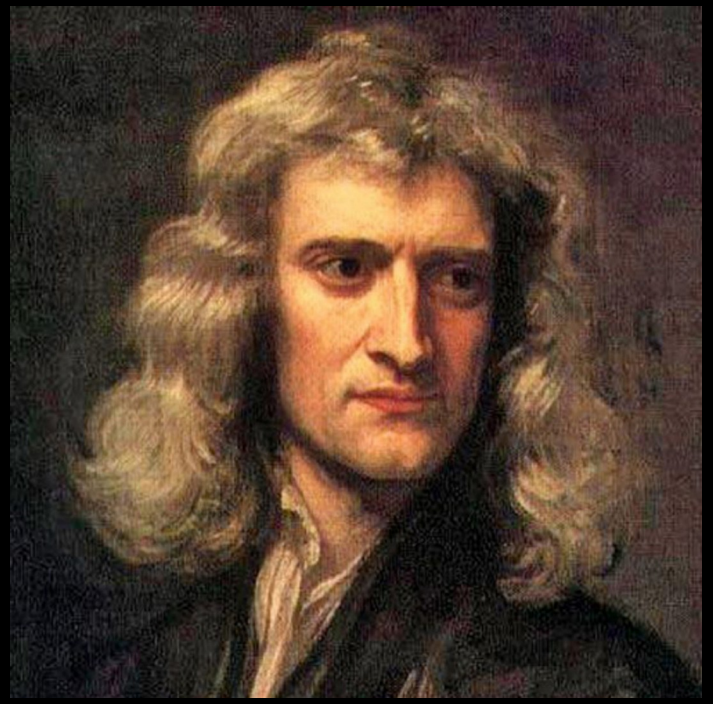
Eratosthenes

measured the Earth, about 240 BC



Johannes Kepler

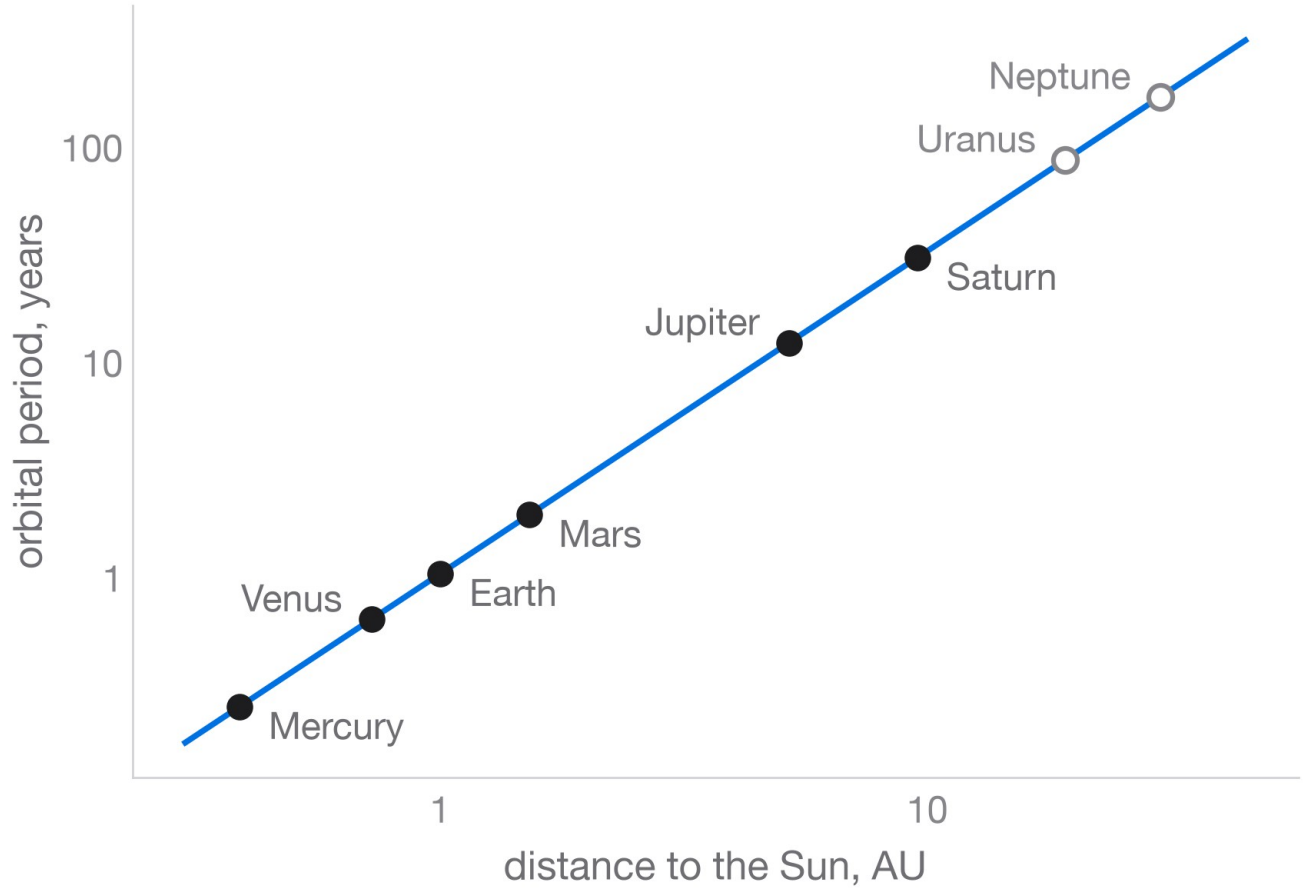
third law of planets, 1619



Isaac Newton

law of gravity, 1687

Kepler's third law, from data



$T = C\ a^p$

p = 1.50

fitted on the six planets Kepler knew. Uranus and Neptune, found later, fit too.

Where math ends and ML begins

1

Theory

Newton derives the law. Data only checks it.

2

Statistics

We pick the form $T = C\,a^{p}$. Data picks p.

3

Machine learning

The form is unknown. We judge a flexible model by its error on new data.

4

Software 2.0

The whole program is learned.

Try it: interactive/02_math_to_ml.html

Data: objects and features

	area, m²	rooms	floor	district	to centre, km	price, $k
flat 1	54	2	3	Chilonzor	6.5	61
flat 2	87	3	9	Mirzo Ulugbek	4.1	118
flat 3	38	1	1	Sergeli	13.0	29
…	…	…	…	…	…	…
new flat	120	4	12	Yunusabad	5.2	?

A row is an object x_i. A column is a feature. The last column is the target y.

Learning means minimising a loss

$\min$ over a: $\quad (1/\ell)\, \Sigma_i\, L(y_i, a(x_i))$

$a(x)$ the model, it predicts $\hat{y}$

$L(y, \hat{y})$ the loss, for example $(y - \hat{y})^2$

ℓ the number of training objects

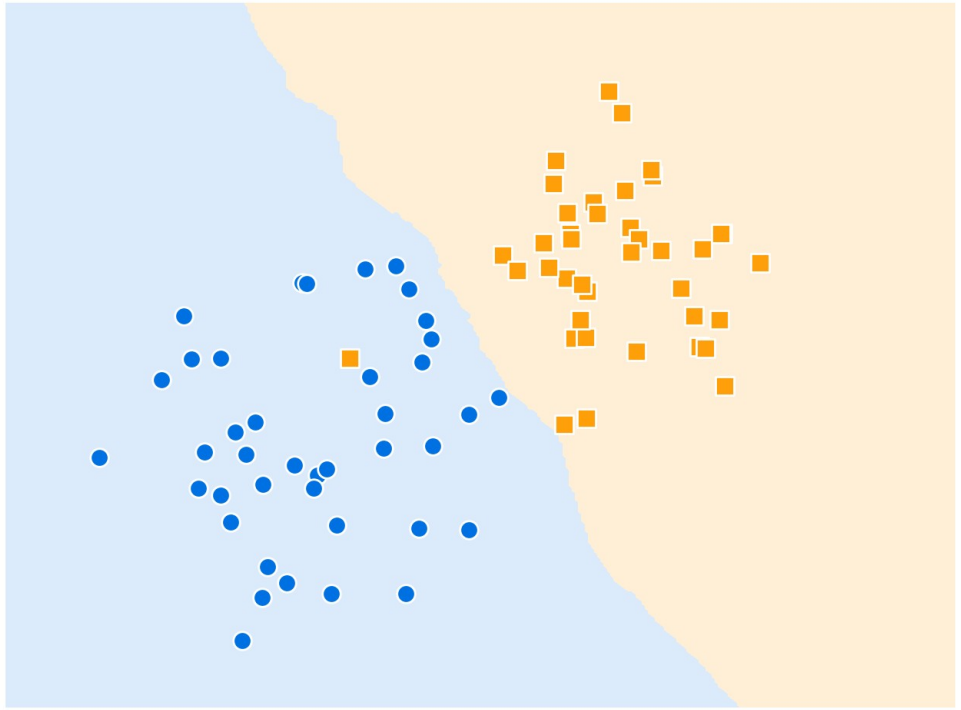
The goal is low loss on new data

Low loss on the training data is easy: memorise it. Doing well on new objects is called generalisation.

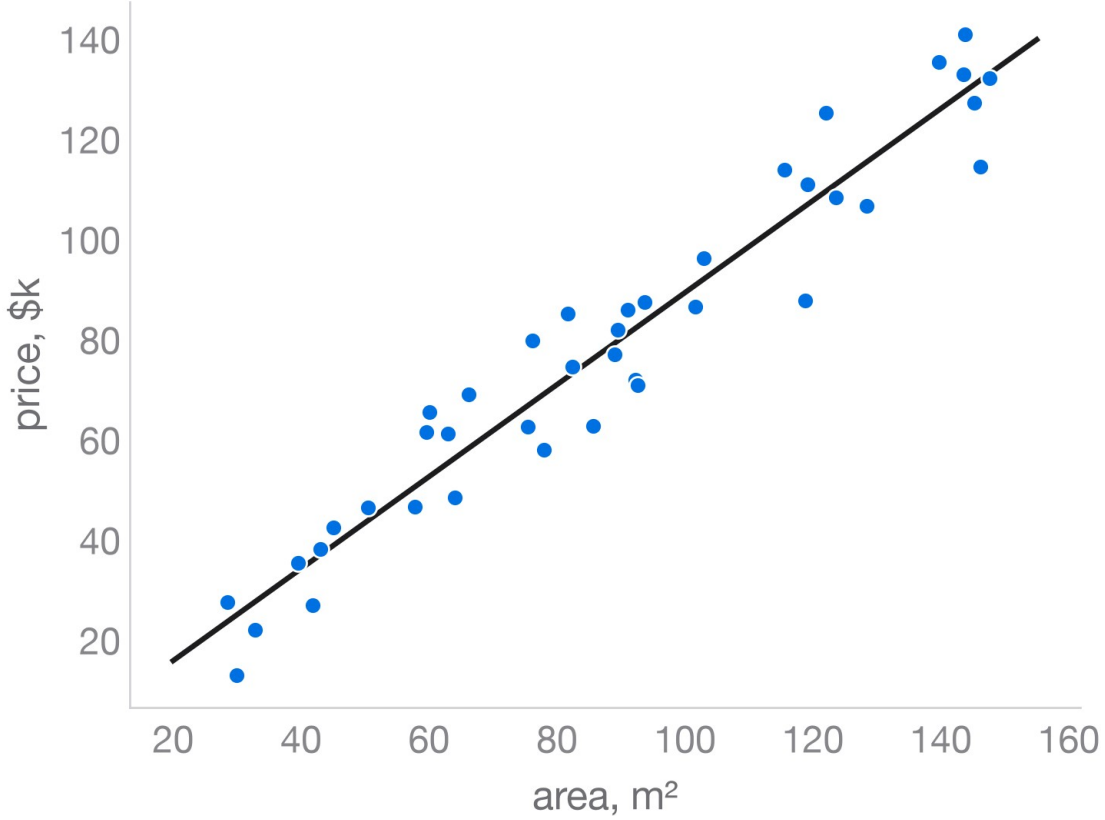
Types of learning

Supervised: objects with answers

Classification: y is a class



Regression: y is a number

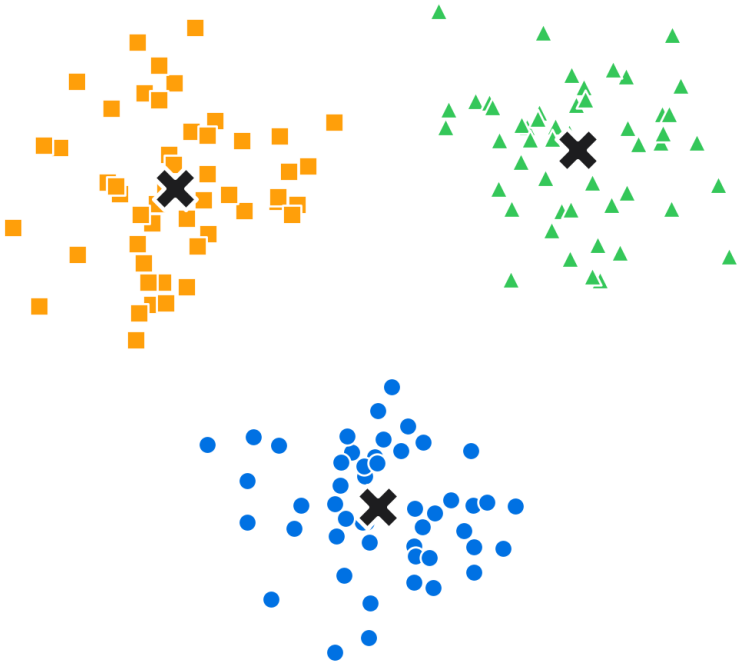


Unsupervised: find structure

No answers, only objects



Three clusters found by k-means



Reinforcement: act, get a reward



Try it: interactive/01_types_of_learning.html

Self-supervised: hide a part, predict it

Plov is the national dish of

Uzbekistan

Raw text supplies the answers for free. This is how large language models are pretrained.

Which type of learning is it?

1. A spam filter trained on emails users marked
2. Flat prices in Tashkent from listings
3. Grouping bank clients into segments

Which type of learning is it?

4 A robot learning to walk

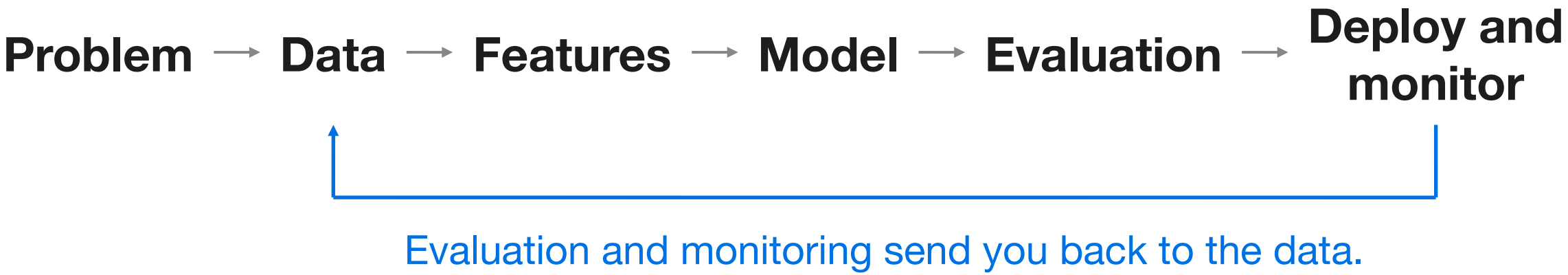
5 Pretraining a language model on raw text

6 Finding fraud without fraud labels

Try it: interactive/01_types_of_learning.html

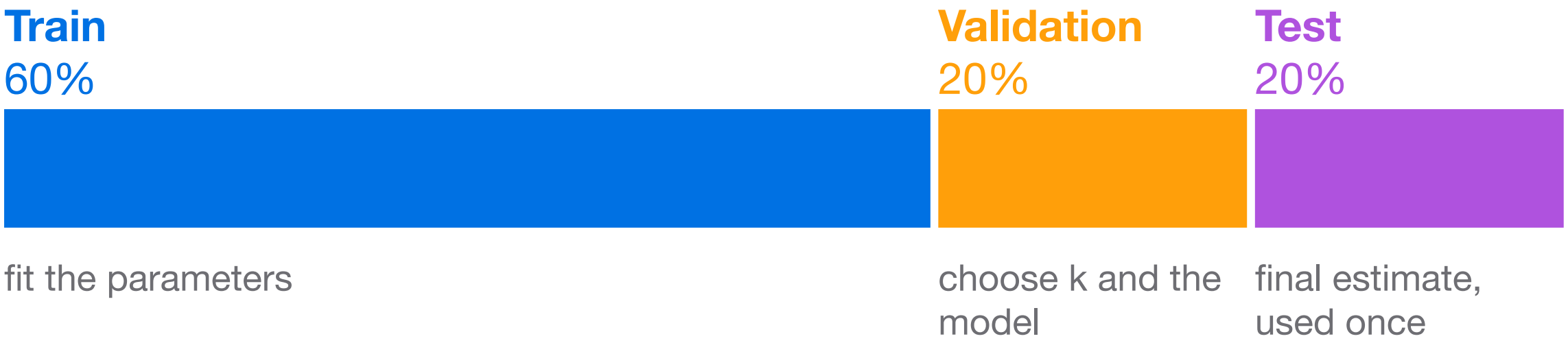
The ML workflow

The ML workflow is a loop



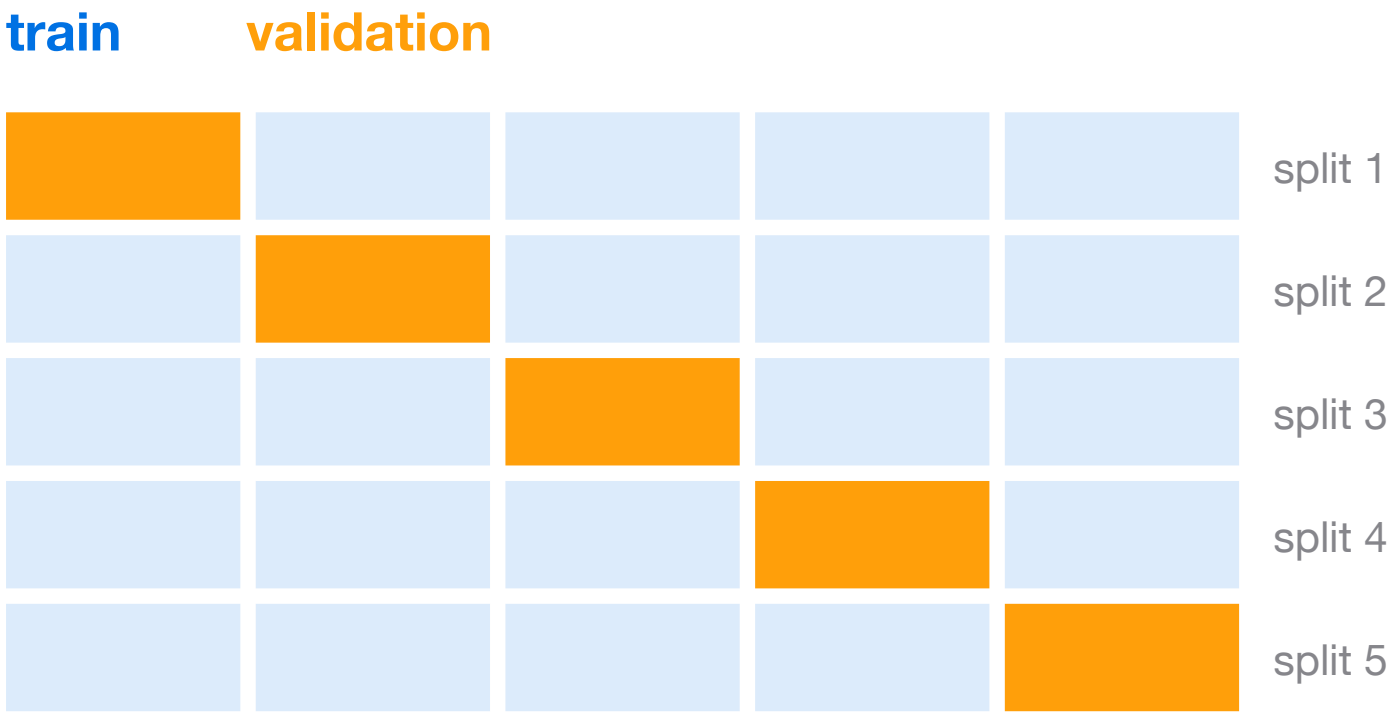
Most of the time goes into the problem, the data and the evaluation.

Train, validation, test



Predicting the future? Split by time.

Small data: cross-validation



Every object is in validation exactly once.

Average the five scores.

Leaks that inflate your score

1. A scaler fitted on all data before the split
2. The same object in train and in test
3. Choosing the model by its test score

Try it: interactive/03_ml_workflow.html

Fit everything on the training part only

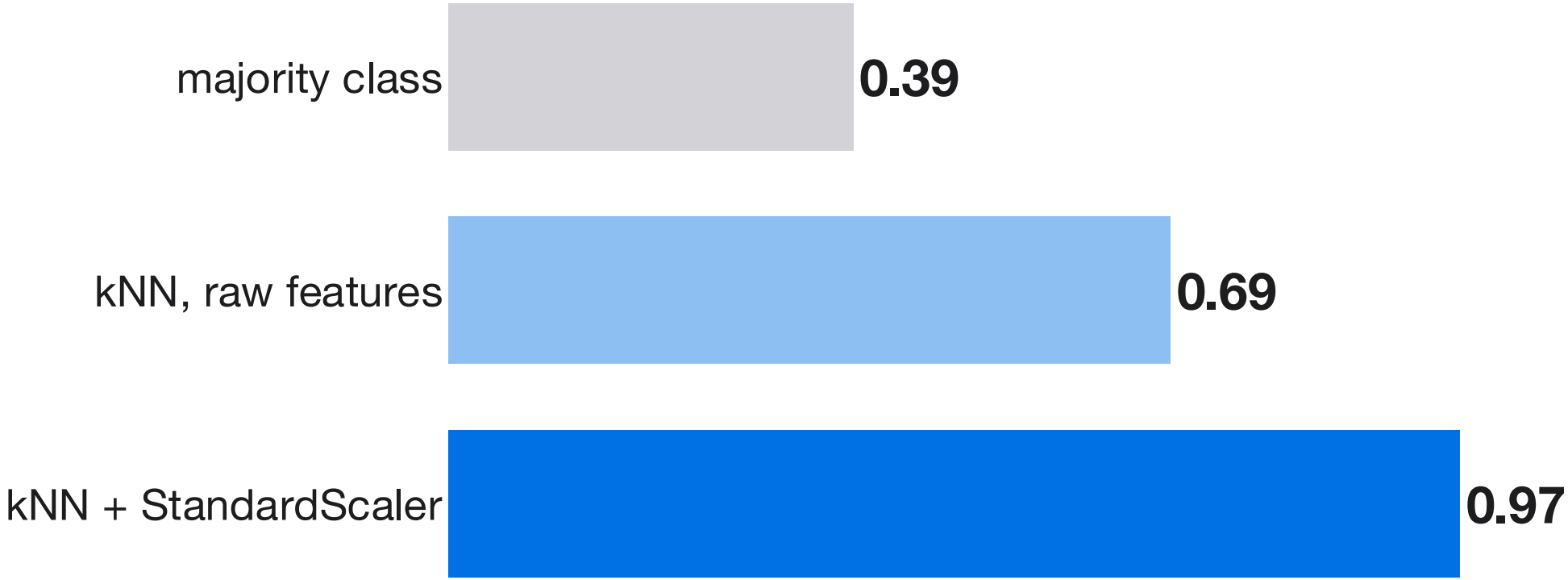
Scalers, encoders, feature selection, k. In scikit-learn, a Pipeline does it for you.

A benchmark is a test set too

Once it is public, it leaks into training data and people tune on it. That is why new benchmarks keep appearing.

Baseline first

Wine dataset, validation accuracy, k = 5



Try it: interactive/03_ml_workflow.html

The Python ML stack

NumPy

arrays and linear algebra

pandas, Polars

tables and EDA

scikit-learn

classic ML, kNN today

matplotlib

plots

PyTorch

deep learning, later

Jupyter, Colab

seminars and homework

Also: git, VS Code, and uv or poetry for environments.

Log every run

run	model	k	scaler	val acc	commit
1	majority		no	0.39	`a1f3c2e`
2	kNN	5	no	0.69	`a1f3c2e`
3	kNN	5	standard	0.97	`b7d901a`
4	kNN	15	standard	1.00	`b7d901a`

You build this log in seminar notebook week0_01_01.

Classic ML and foundation models

Foundation models

Pretrained on huge, broad data, mostly self-supervised

Adapted by prompting, fine-tuning, or used for embeddings

GPT, Claude, Gemini, Qwen, Llama; CLIP for images; Whisper for speech

Open models are close behind

4 months

Open-weight models trail the best closed ones by about four months.

Epoch AI, May 2026

46 vs 60

Best open model (Kimi K3.1 Pro) against the best closed one (Gemini 4 Carbon).

Artificial Analysis Intelligence Index, November 2026

Open-weight leaders: Kimi, DeepSeek, GLM, Qwen.

Which one wins?

Classic ML

Tables with many labelled rows

Tight latency, cost or regulation

Runs on your servers or a phone

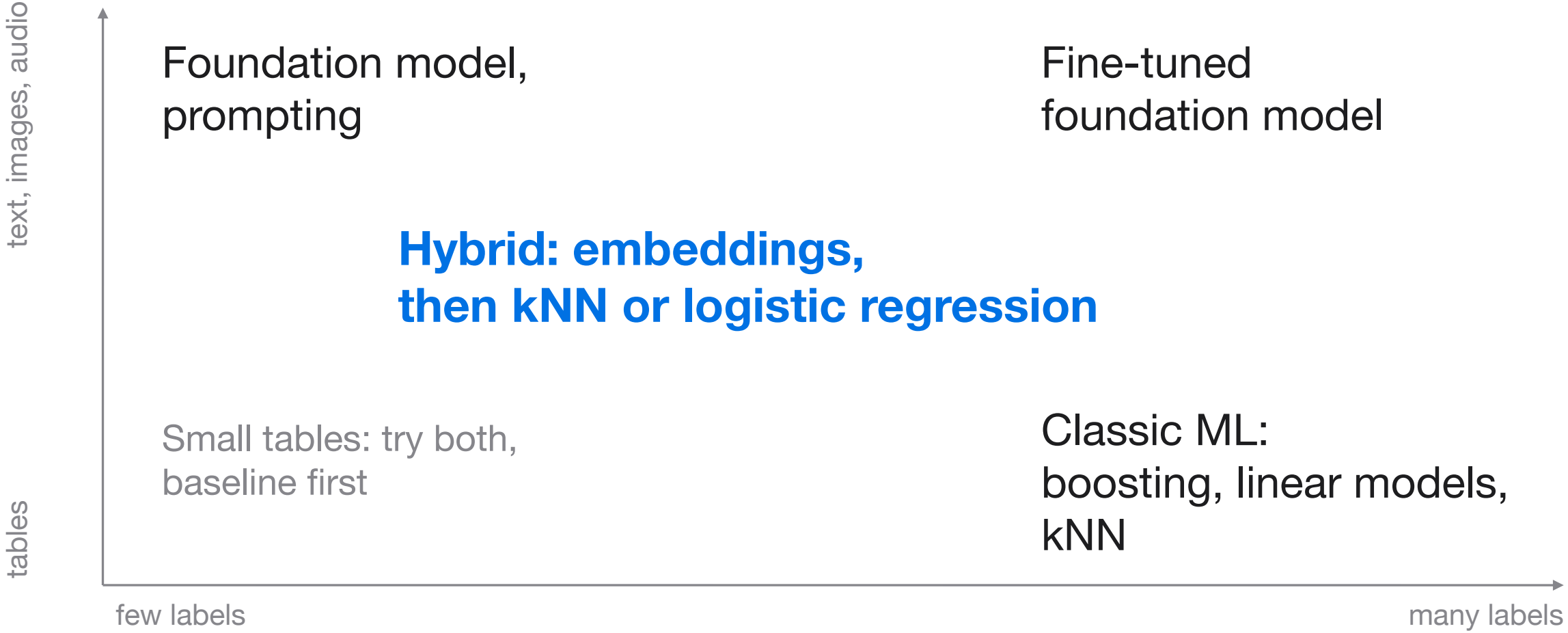
Foundation models

Text, images, audio

Few or no labels

Generation, world knowledge, many languages

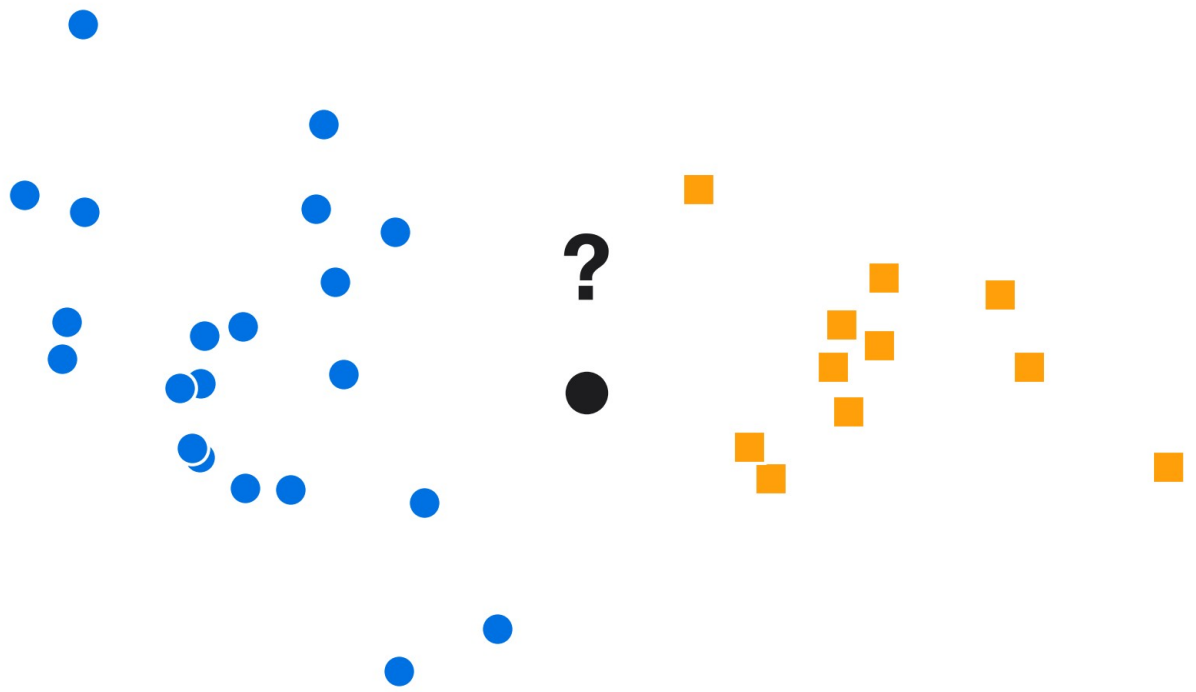
Hybrid: embed, then kNN



Try it: interactive/04_classic_vs_foundation.html

k nearest neighbours

Which class is the new point?

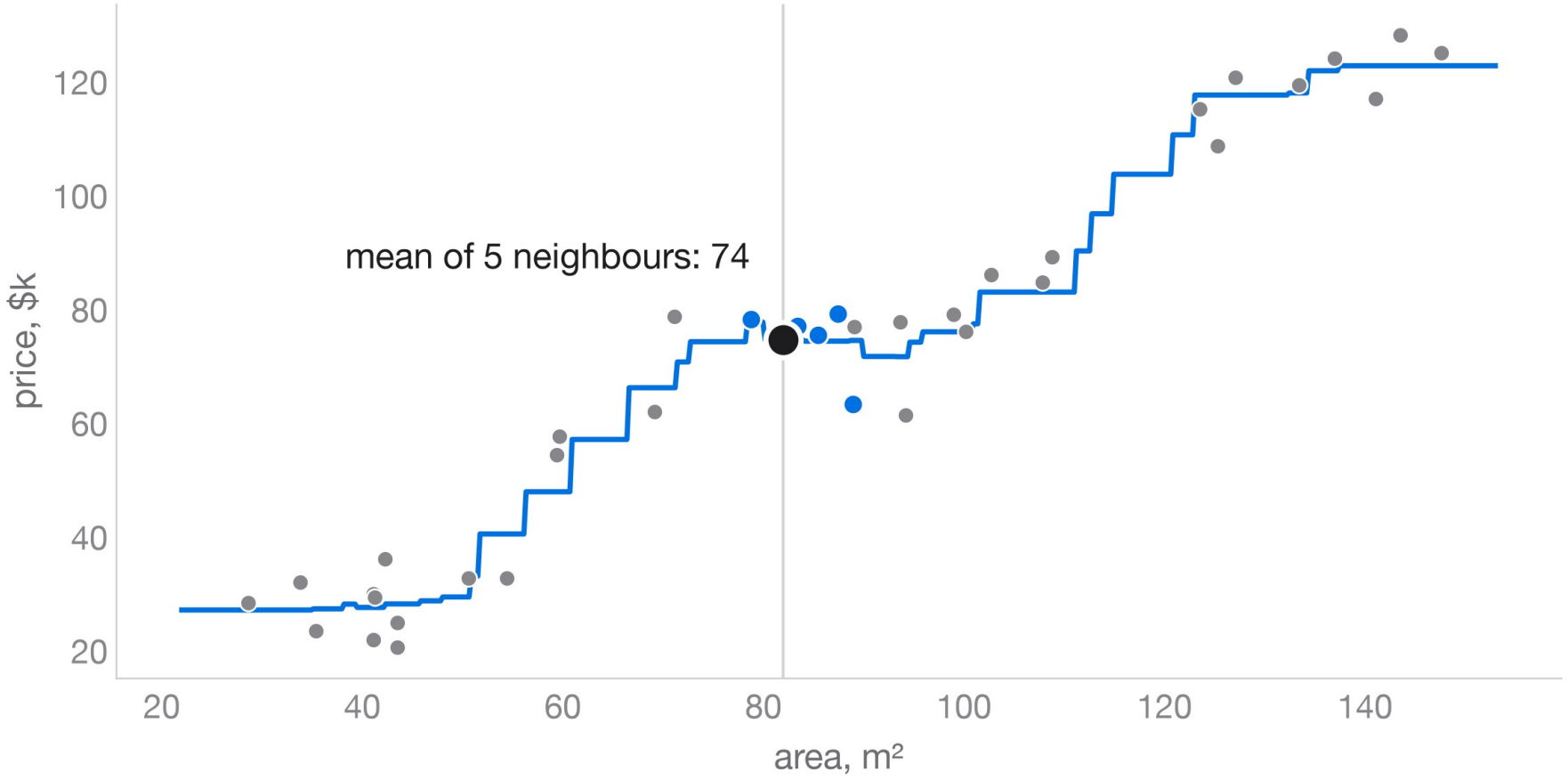


kNN in three steps

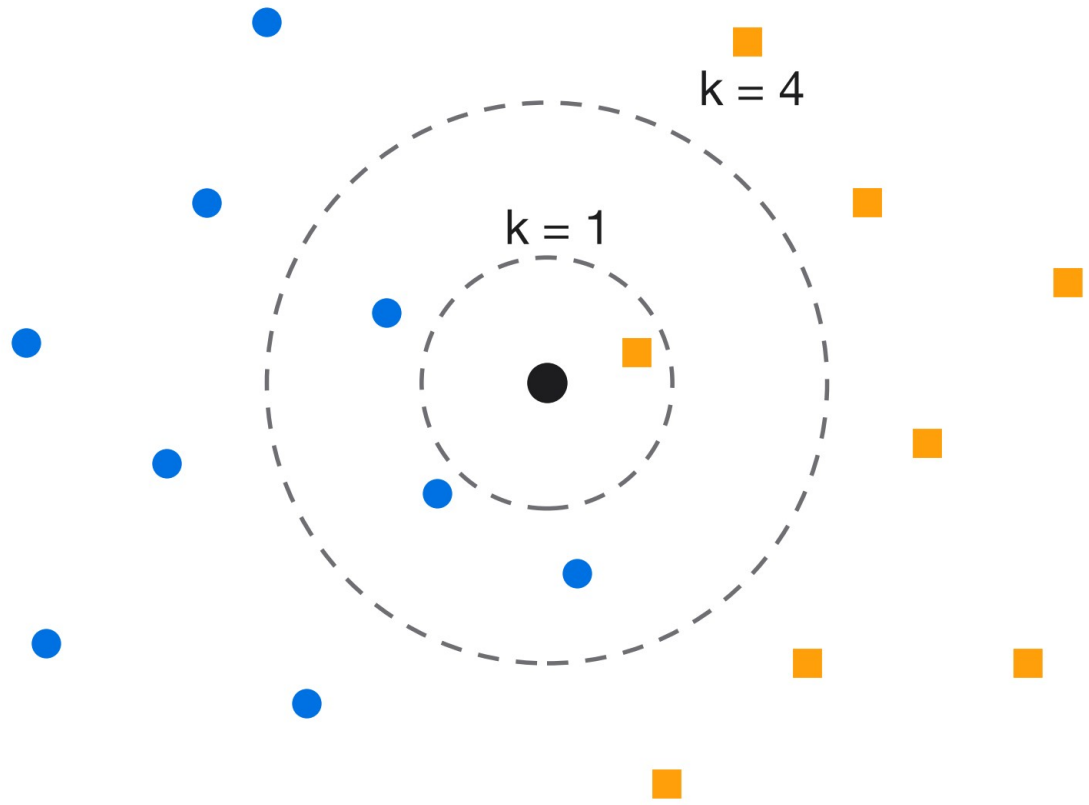
1. Measure the distance to every training object
2. Take the k closest
3. Vote for a class, or average the numbers

Try it: interactive/05_knn_playground.html

Regression: average the neighbours



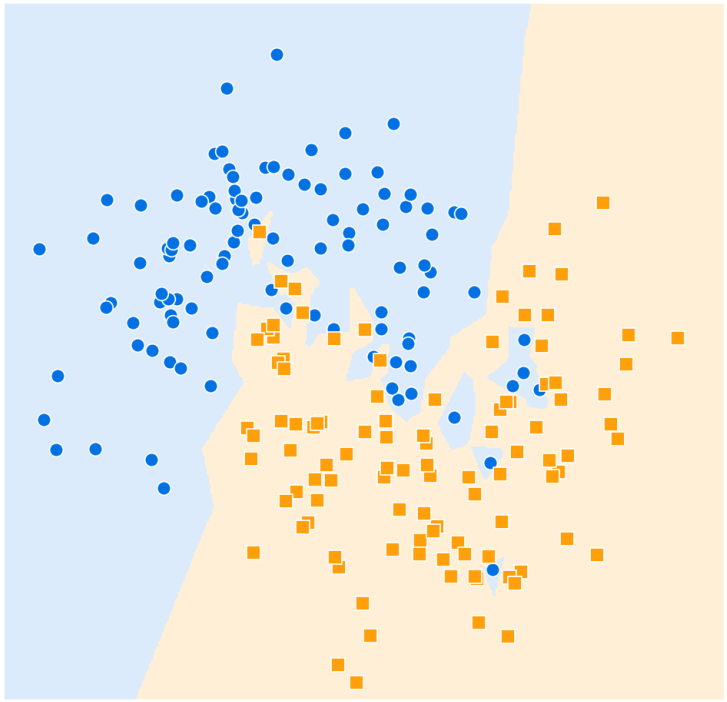
The answer depends on k



k = 1: square

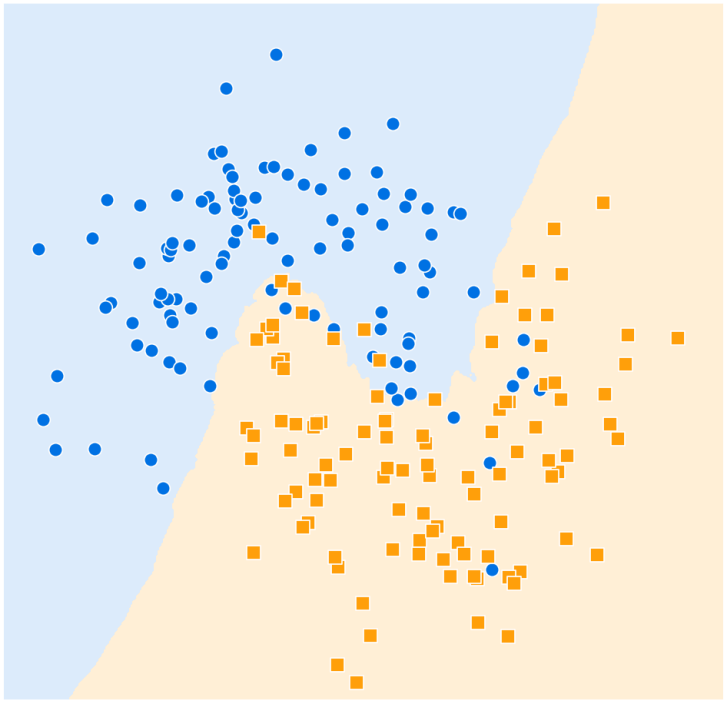
k = 4: circle

Small k overfits, large k underfits



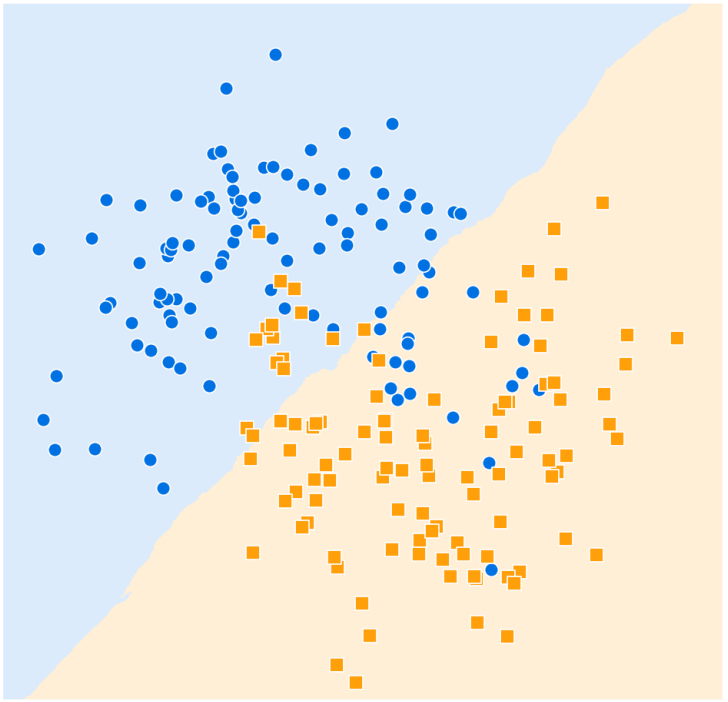
k = 1

train 1.00 test 0.90



k = 15

train 0.91 test 0.93



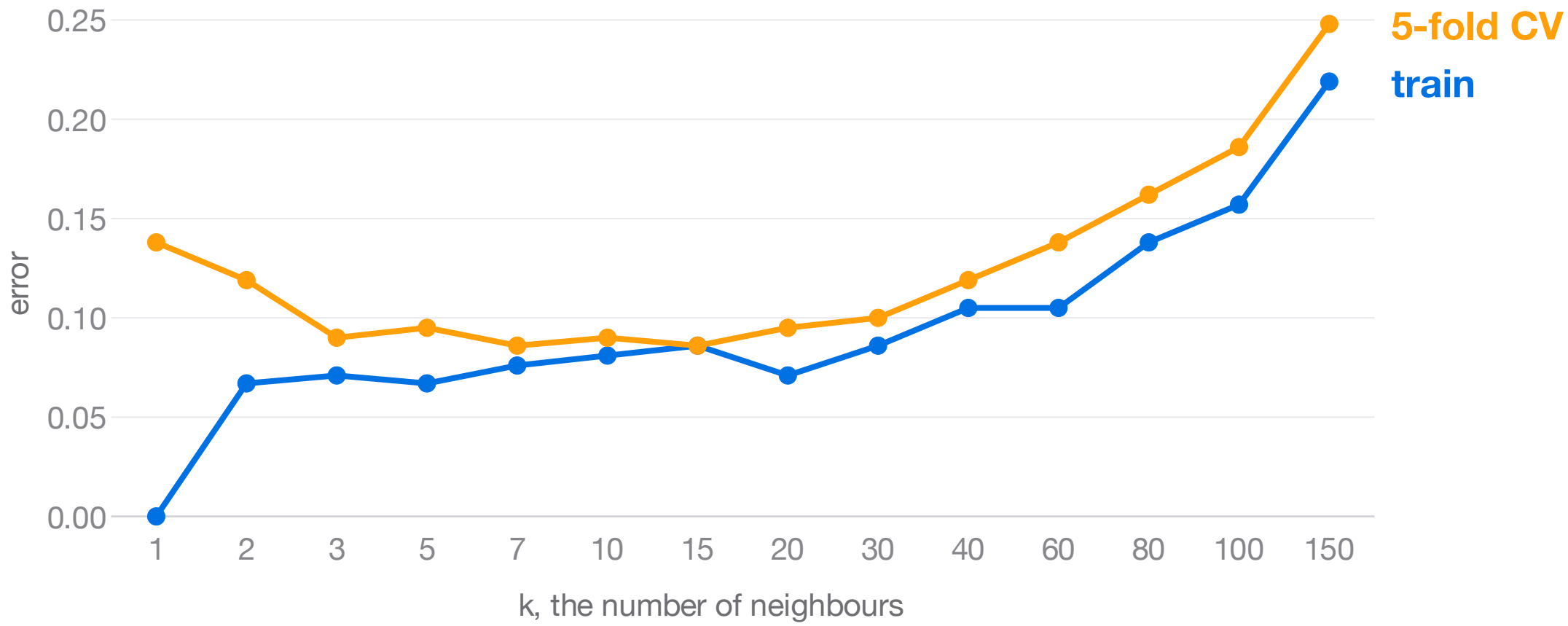
k = 100

train 0.84 test 0.76

Try it: interactive/05_knn_playground.html

Choose k by cross-validation

Noisy two moons, 210 training points

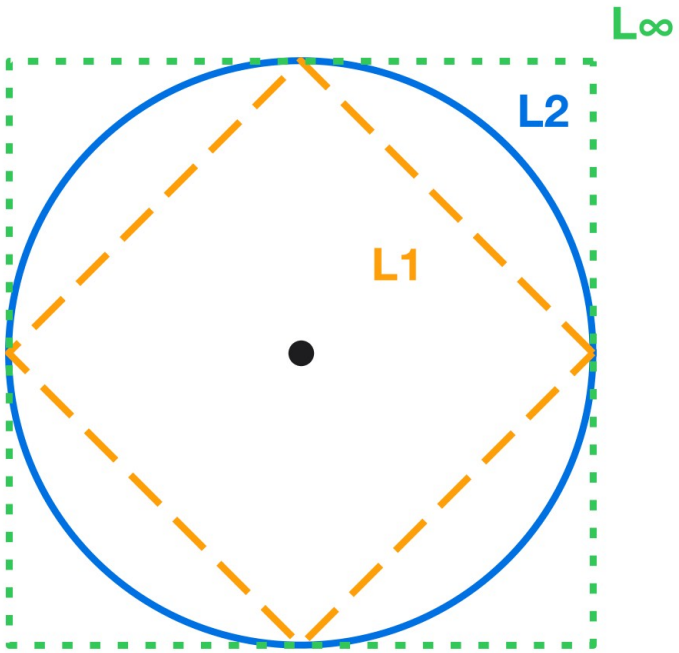


Try it: interactive/06_knn_choosing_k.html

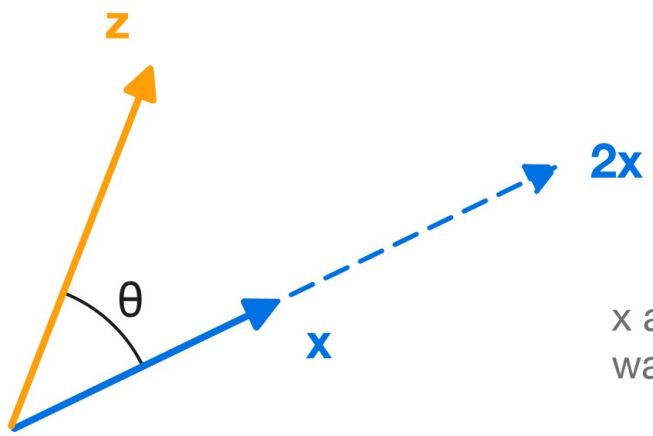
The metric defines “close”

Cosine for texts and embeddings. Hamming for strings and categories.

Points at distance 1



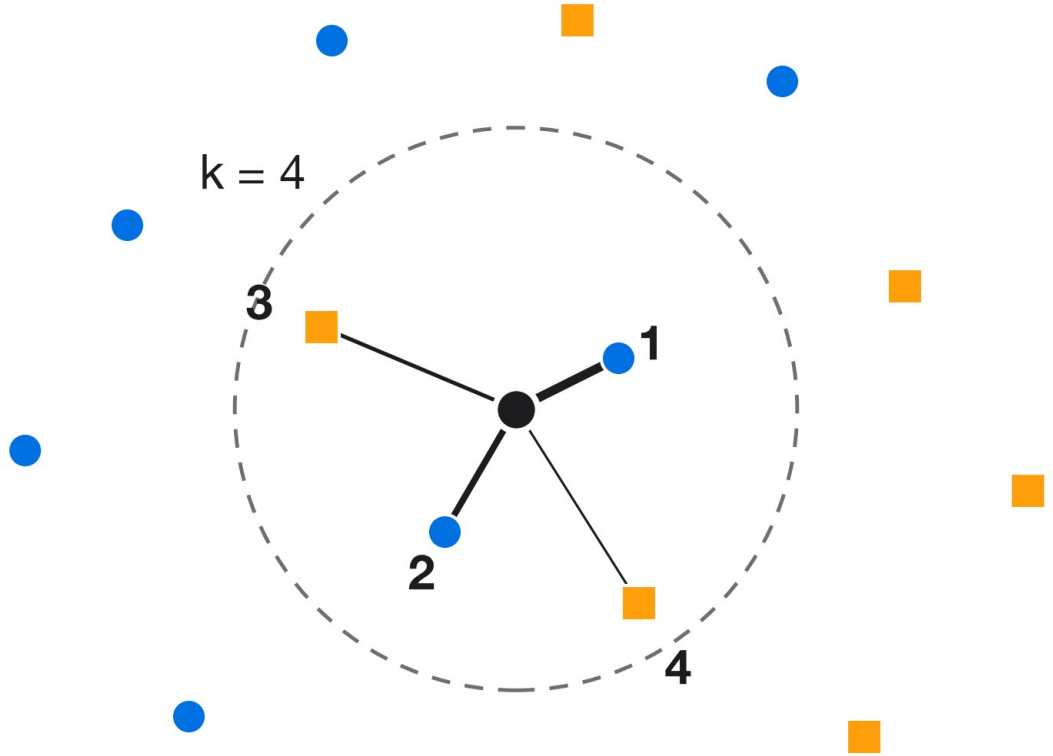
Cosine: only the angle counts



x and 2x point the same way: cosine distance 0

Try it: interactive/05_knn_playground.html

Closer neighbours get more weight



By rank: $w_{(i)} = w_i$

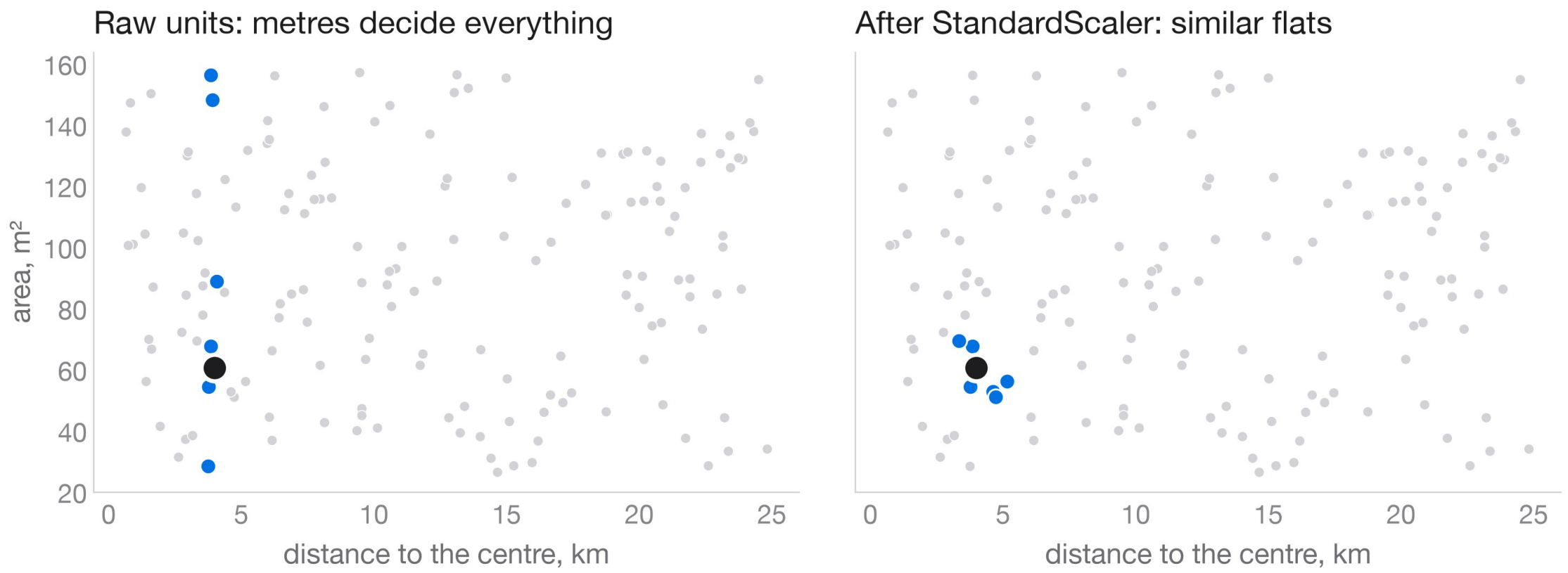
By distance: $w_{(i)} = w(d(x, x_{(i)}))$

p(circle) = $(w_1 + w_2) / \Sigma w_i$

Without weights it is a 2 to 2 tie.

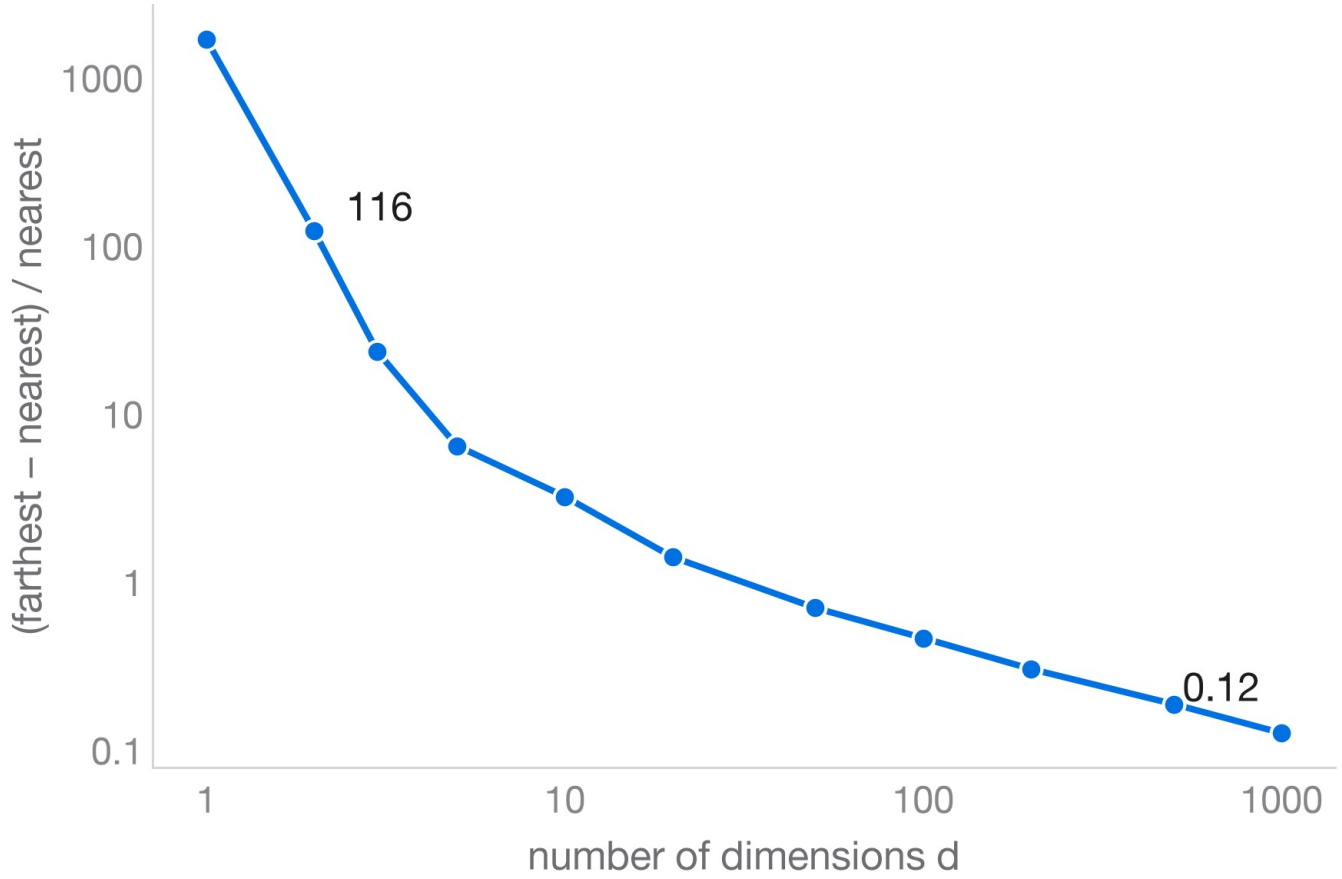
Scale the features

Flat prices from last year's lecture 4: area in m², distance in metres.



Try it: interactive/07_feature_scaling.html

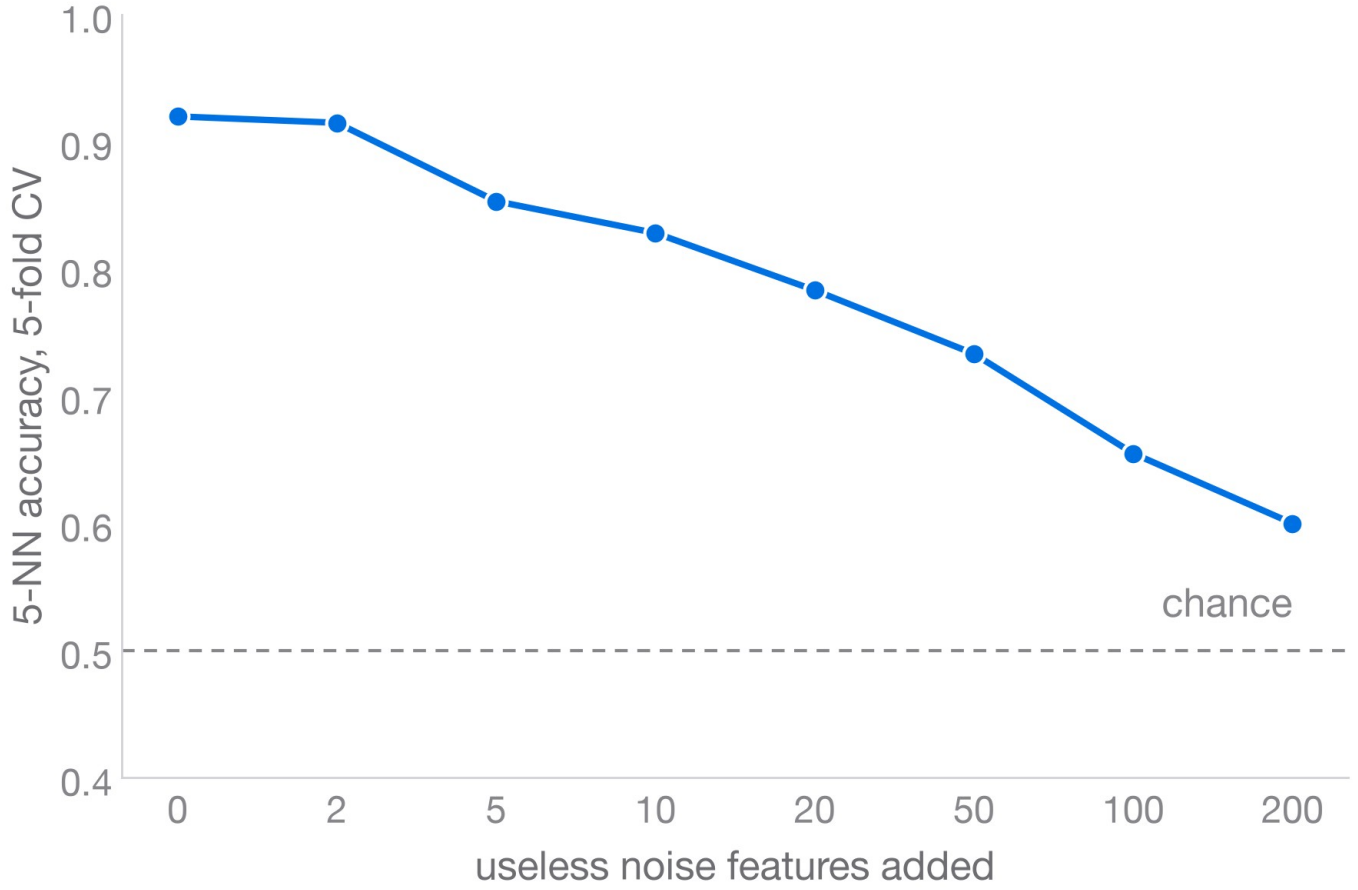
High dimensions: everything is far



0.12

In 1000 dimensions the farthest point is only 12% farther than the nearest.

Useless features hurt kNN



0.60

accuracy with 200 noise features, down from 0.92

Try it: interactive/08_curse_of_dimensionality.html

0.1 s

to find 10 nearest neighbours by brute force

10^6 vectors of 100 numbers, 10^9 operations per second. Fine once, too slow for thousands of queries a second.

A kNN index makes search fast

Build it once, then search. Approximate search trades a little accuracy for a lot of speed.

Trees

k-d tree and ball tree, in scikit-learn

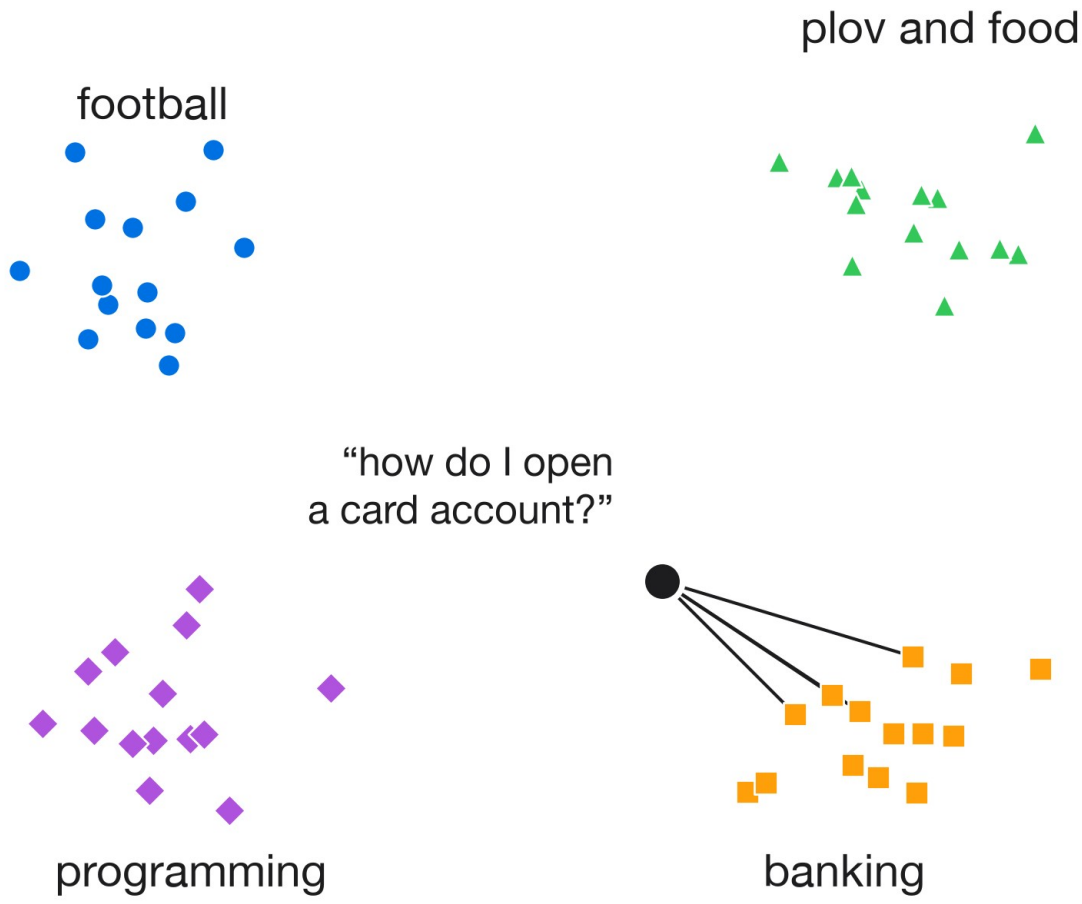
Clusters

IVF: search only the nearest clusters

Graphs

HNSW, the usual choice today

kNN today: search in embedding space



Semantic search and RAG

Recommendations

Face recognition and deduplication

Try it: interactive/09_knn_and_embeddings.html

kNN: pros and cons

Pros

No training: just add data

Classes and numbers alike

Easy to explain: “because of these neighbours”

Cons

Slow search without an index

Needs scaled, relevant features

Weak in high dimensions

Revise

ML learns rules from data. We judge it on new data.

Split train, validation, test. No leaks. Baseline first.

kNN: pick k and the metric by validation. Scale features.

Exit quiz

Scan the code.

Enter your full name. One question at a time, 30 seconds each.

If time allows, we then play the multiplayer game Neighbours.



165.22.93.84

Practice and homework

Seminar

Two notebooks: workflow and baseline, then kNN practice

Interactive

Nine pages: play with k, metrics and scaling at home

Homework 1

kNN from scratch in NumPy: two loops, one loop, no loops

Interactive pages: 165.22.93.84/viz. Notebooks and homework: links in the course chat.

Next time: Naive Bayes and maximum likelihood

Also: where to find datasets and how to explore them. Bring your questions about homework 1.

Questions?

Makhmudjon Sodikov, YMLS Uzbekistan

YMLS



girafe ai